What's wrong with being in a false vacuum? Quite possibly, everything. A false vacuum is at best a temporary reprieve from ultimate destruction. In a false vacuum, the laws of physics, including the ability of particles to exist at all, are contingent on a precarious balancing act that could be upset at any moment.

When this happens, it's called vacuum decay. It's quick, clean, painless, and capable of destroying absolutely everything.

A BUBBLE OF QUANTUM DEATH

In order for vacuum decay to occur, there has to be a trigger—something that will set the Higgs field wandering far enough to find the part of the potential corresponding to the "true" vacuum and realize it would rather be there.[fn8] An ultra-high-energy explosion, or the catastrophic final evaporation of a black hole, or even an unfortunate quantum tunneling event (more on these later) could set it off. If this happens anywhere in the cosmos, it creates an unstoppable apocalyptic cascade that nothing in the universe can withstand.

It starts with a bubble.

Wherever the event occurs, a tiny bubble of true vacuum forms. This bubble contains within it a drastically different kind of space—one in which the processes of physics follow different laws, and the particles of nature are rearranged. At the moment it forms, it's an infinitesimal speck. But it is already surrounded by a bubble wall of extremely high energy that could incinerate anything it touches.

Then, the bubble begins to expand.

Because the true vacuum is the more stable state, the universe "prefers" it, and will revert to it if given the slightest chance, just like a pebble will roll down a slope if it's placed on one. As soon as the bubble appears, the Higgs field all around it is suddenly being shaken down to the valley floor. It's as though that first event knocks free every precariously balanced pebble near it, and then the avalanche spreads. More and more space succumbs to the true vacuum state. Anything unfortunate enough to be in the bubble's path is first hit by the intensely energetic bubble wall, approaching at about the speed of light. Then it undergoes a process that could only be called total and complete dissociation, as the forces that